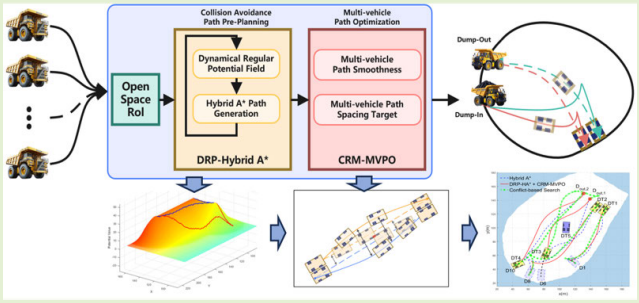
Collision Risk-Minimized Multivehicles' Cooperative Path Planning in Open-Pit Mine Unloading Areas

Yunchao Chen, Desheng Zhu, Xiaotian Ke, Kehu Yang, *Member, IEEE*, Jianjian Yang, and Shirong Ge

***Abstract*—Multivehicle collaborative trajectory planning for autonomous vehicles has attracted significant attention, yet research aimed at reducing spatial collision probability remains limited. In open-pit mine unloading areas, with their complex environment, flexible tasks, and inertia-induced motion delays in dump trucks, traditional multivehicle planning methods based on spatiotemporal trajectories produce planned results that impose strict timing and tracking requirements. This increases the difficulty of controller execution and substantially preserves the risk of vehicle collisions. To address these challenges and fully exploit the open space of unloading areas, this article innovatively proposes a spatial avoidance-based cooperative multivehicle path planning method to minimize collision risks in unstructured environments such as unloading areas. An improved Hybrid A* algorithm is proposed, incorporating a hybrid A* incorporating dynamic routing potential fields (DRP-Hybrid A*) to iteratively generate coarse solutions for the dispersion of multivehicle paths. Furthermore, a collision risk minimization multivehicle path optimization (CRM-MVPO) algorithm is proposed, which optimizes the path through smoothness and safety distance, minimizing discontinuities and collision risks. Simulation experiments are conducted under two categories, comprising three typical scenarios to validate the approach's effectiveness: single-entry–single-exit and multiple-entry–multiple-exit multi-dump-point configurations. The experimental results demonstrate that the proposed method effectively reduces the probability of path overlap and collision risks, enhancing the safety and flexibility of multivehicle cooperative operations, and providing a reliable and efficient solution for autonomous driving in complex mining areas.**



***Index Terms*— Autonomous driving, multivehicle cooperation, open-pit mining, path optimization, path planning.**

I. Introduction

With the growing demand for mineral resources, autonomous transportation systems in open-pit mines are gaining attention for their ability to reduce labor costs, improve efficiency, and enhance safety [1], [2].

Advanced sensor technologies, such as LiDAR and GNSS, equip autonomous mining trucks (AMTs) with precise, real-time data, enabling safe navigation in the complex and unstructured environments of open-pit mines [3], [4], [5]. The integration of these sensors allows AMTs to perceive their surroundings with greater accuracy, forming a critical foundation for trajectory planning [6]. However, the unstructured nature of open-pit mine unloading areas, characterized by irregular boundaries, frequent environmental changes, and the simultaneous demands of unloading operations and collision avoidance, considerably complicates path planning [7]. Moreover, multiple AMTs are typically deployed concurrently to enhance operational efficiency, increasing the likelihood of path conflicts and access competition. For instance, the trajectory of an earlier-arriving truck may obstruct subsequent vehicles, intensifying conflicts and potentially leading to congestion.

To address these challenges, this study proposes a multivehicle cooperative path planning method to minimize collision risks in unstructured environments like open-pit mine unloading areas, aiming to improve safety and operational flexibility.

A. Motivation

Traditional planning strategies primarily focus on single-vehicle paths, but, in multivehicle environments, path con-

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flicts among AMTs significantly increase the complexity of the problem. Existing multivehicle path planning (MVPP) methods—such as conflict-based search (CBS) [8], spatiotemporal sampling [9], and spatiotemporal corridor planning [10], [11]—often produce plans that demand precise timing coordination and trajectory tracking for successful execution. These demands can complicate controller execution and heighten collision risks under disturbances. There are also approaches based on first-in–first-out (FIFO) rules and iterative computation-based planning methods [12]. However, these approaches fail to adequately address the specific challenges of open scenarios, such as unloading areas, where the sparse environment could be leveraged to reduce spatial conflicts.

Few studies address the multivehicle path optimization problem, with most focusing on single-path optimization. Approaches such as H-OBCA [13], [14] and DL-IAPS [15] optimize the smoothness of individual vehicle paths while avoiding collision constraints or employing local smoothing techniques and spline curves to ensure the continuity and feasibility of the path. However, these methods primarily target single-path trajectory/path optimization and overlook vehicle interactions in multivehicle systems. Achieving smooth transitions and coordination in dynamic, unstructured environments requires more precise handling of path conflicts and interaction effects between vehicles.

The unloading area is a more plentiful space, and for multivehicle path planning, current methods often fail to utilize the open area fully, as shown in Fig. 1. Therefore, a path design that is as decentralized and safe as possible can significantly reduce potential conflicts between vehicles, relax the constraints on traffic order, and reduce the risk of deadlock.

B. Contributions

To address path conflicts in multivehicle operations within open-pit mine unloading areas, we propose a cooperative path planning method to minimize collision risks. The method includes three core steps.

1) *Dynamic Routing Potential Field Model:* A dynamic potential field [16] model is created by calculating the path potential between vehicles and adjacent dump points. This model prevents conflicts and guides vehicle paths, improving flexibility and adaptability.
2) *Path Search Method Incorporating Dynamic Routing Potential Field:* A hybrid A* incorporating dynamic routing potential fields (DRP-Hybrid A*) algorithm is proposed to generate initial paths quickly. It integrates the dynamic potential field into the Hybrid A* algorithm, ensuring path safety, improving path decentralization, and multivehicle path planning efficiency.
3) *Collision Risk Minimization Multivehicle Path Optimization Algorithm (CRM-MVPO):* To enhance path smoothness and minimize intervehicle collision risk, we propose the CRM-MVPO algorithm, which refines preliminary paths by considering vehicle kinematics and intervehicle safety spacing. The method reduces abrupt path changes and redundancy while maintaining safe separation between vehicles, ensuring both feasibility and collision avoidance in multivehicle scenarios.

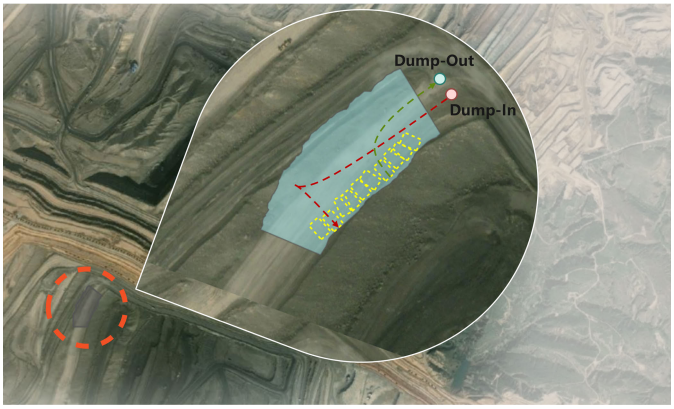


Fig. 1. Unloading area of open-pit mine.

C. Organization

The remainder of this study is organized as follows. Section II presents in detail the work related to the MVPP method. Section III describes the MVPP problem for unstructured unloading areas in mathematical language and presents the framework of this work. Section IV describes the details of the DRP-Hybrid A* path planning method and the CRM-MVPO method. Section V presents the results of three simulation experiments. Finally, Section VI provides a summary.

II. Related Works

Minimizing collision risk is a core challenge in multivehicle cooperative path planning for unstructured scenarios. Ensuring both the safety and operational efficiency of each vehicle in dynamic environments has become a key research focus in the field of path planning.

In recent years, numerous improved algorithms based on A* [17] or Hybrid A* [18] have been proposed for single-vehicle path planning to address collision avoidance tasks in diverse environments. For example, Shan et al. [19] combined artificial potential fields with A* by incorporating the potential field into its heuristic function. This method searches for a path that avoids obstacles as much as possible, preventing collisions, but does not consider vehicle kinematics. Li et al. [20] proposed the global optimal enhanced Hybrid A* (GHA) algorithm, which uses a greedy strategy to improve global optimality. This method finds optimal paths under complex constraints and is well-suited for dynamic environments.

Compared to single-vehicle collision avoidance path planning, multivehicle path planning significantly increases the problem complexity [21], [22]. Multivehicle path planning can be understood as each vehicle planning its path while treating other vehicles as dynamic obstacles and simultaneously ensuring the feasibility and optimality of its path [23], [24]. Therefore, the paths must avoid conflicts with other vehicles while ensuring smoothness and dynamic feasibility, thereby satisfying the overall system performance requirements.

For multivehicle collision avoidance planning, Sharon et al. [8] proposed the CBS algorithm with a

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two-layer framework. The upper layer detects conflicts and creates constraints, while the lower layer uses A* to find collision-free paths. The layers work together iteratively to ensure conflict-free paths. Building on this, Wen et al. [25] replaced the lower-level planner in the CBS framework with a Hybrid A* algorithm to coordinate car-like robots. However, frequent robot-to-robot collisions result in exponential growth of the constraint tree.

Several improved Hybrid A* algorithms consider time and vehicle speed. For example, the 4-D V-Hybrid A* algorithm [26] generates initial trajectory estimates for path optimization, while time Hybrid A* (THA*) [27] incorporates steering, acceleration, and collision avoidance to resolve conflicts. However, these algorithms are sensitive to time resolution, with large resolutions causing search failures.

The multivehicle trajectory planning problem can be described as an optimal control problem (OCP) and discretized into a nonlinear programming (NLP) [28] problem. Common approaches include mixed-integer programming (MIP) [29], mixed-integer linear programming (MILP) [30], mixed-integer nonlinear programming (MINLP) [20], mixed-integer quadratic programming (MIQP) [31], sequential convex programming (SCP) [32], and incremental sequential convex programming (iSCP) [33]. Hult et al. [29] addressed multivehicle task allocation using mixed-integer programming, offering a low complexity approach for central coordination and local control. Augugliaro et al. [32] applied SCP, linearizing nonlinear collision avoidance constraints and solving iteratively. Chen et al. [33] proposed iSCP, which updates constraints incrementally but does not guarantee the feasibility of the intermediate solution. Li et al. [34] introduced adaptive-scaling constrained optimization (ASCO) to optimize the MVTP problem. Ouyang et al. [35] combined global and local optimization in a nonconvex static obstacle environment, first identifying a feasible homotopy class and then finding a local optimum. However, these methods face challenges in practical applications, especially as the number of vehicles increases and the environment becomes more dynamic. Most optimization methods also rely on a good initial solution to speed up the solving process.

Path smoothness is critical for path planning quality, with nonlinear constraints like curvature affecting optimization efficiency. Zhang et al. [13] proposed a hierarchical optimization-based collision avoidance (H-OBCA) method integrating vehicle dynamics and collision constraints within a convex framework to ensure smooth, feasible paths. He et al. [36] introduced a time-partitioned distributed optimization (TPD-OBCA) that reduces computational complexity via iterative subproblem solving. However, both focus on single-path optimization and do not explicitly address multivehicle smoothness or collision risk. In practice, Baidu Apollo's FEM POS deviation method smooths reference lines by solving a quadratic program balancing position deviation, curvature, and smoothness. Likewise, B-splines, Bézier curves, and polynomial fitting offer flexible smooth path generation but lack consideration of multivehicle interactions in dynamic settings.

Therefore, the core challenge in multivehicle collision avoidance and cooperative path planning is minimizing collision risks while optimizing path smoothness and dispersion. This requires integrating dynamic routing and global coordination mechanisms into the path planning framework to achieve efficient and robust results in complex environments.

III. Problem Statement and Framework

A. Problem Statement

Given N AMTs operating concurrently in the unloading area of an open-pit mine, the boundary set of this area is denoted as $\mathcal{R} = \{b_1, b_2, \ldots, b_n\}$, where each $b_i \in \mathbb{R}^2$ represents the geographic coordinates of a boundary point. As illustrated in Fig. 2, the unloading area includes an entrance D_{in}, an exit D_{out}, and a set of task points $\{D_1, D_2, \ldots, D_M\}$, where each $D_i \in \mathbb{R}^2 \times \mathbb{S}^1$ represents the position and orientation of a task location. For each mining truck, its operational workflow begins with entering the area via D_{in}, proceeding to a designated unloading point D_n to complete the unloading operation, and finally exiting through D_{out}. To this end, a planning algorithm is required to derive the corresponding driving paths. The path to the nth unloading point is denoted as P_n

$$P_n = \{(x_n(i),\ y_n(i),\ \theta_n(i)) \,|\, i = 0, 1, \ldots, n_e\} \tag{1}$$

where $(x_n(i),\ y_n(i),\ \theta_n(i))$ represent the spatial coordinates and heading angle at the ith point, with $\theta_n(i) \in [-\pi, \pi]$. These points are obtained by uniformly sampling with a fixed interval Δs along the path, and n_e is the total number of sampled points.

There exists a potential risk of crossovers between the incoming and outgoing truck trajectories. To ensure safe and efficient operation, the objective is to minimize the overlap regions between different routes and path roughness, formalized as

$$\min \sum_{\substack{i,j\in\{1,\ldots,N\}\\ i\neq j}} J_{\text{spacing}}\left(P_i, P_j\right) + \sum_{n=1}^{N} J_{\text{rough}}\left(P_n\right) \tag{2}$$

where J_{spacing} penalizes the spatial proximity between the paths of different trucks and $J_{\text{rough}}(P_n)$ evaluates the smoothness of the nth truck's path.

B. Framework

Fig. 3 illustrates the overall framework of our approach, which consists of two core modules: a Hybrid A* iterative multivehicle path planning method incorporating dynamic routing potential fields (DRP-Hybrid A*) and a collision risk minimization multivehicle path optimization algorithm (CRM-MVPO).

Since it is not feasible to frequently adjust truck paths in real open-pit mines, the first module aims to preplan as decentralized or even completely collision-free truck paths as possible. The initial paths generated by DRP-Hybrid A*, although satisfying all constraints, may be suboptimal and need to be further optimized. This is because of the following.

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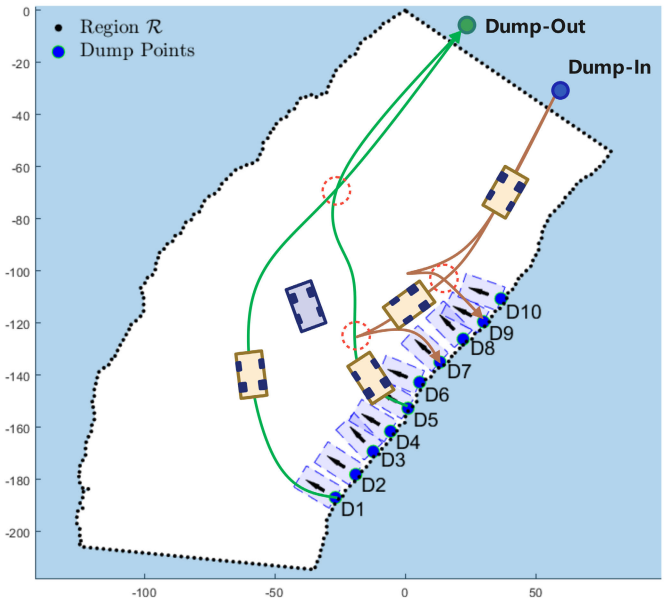


Fig. 2. Path conflicts in unloading area: light blue dashed boxes labeled D_1–D_{10} indicate selectable dump points, and red dashed circles mark collision risk critical points.

1) The curvature profile of the initial path may have discontinuities that, if used directly in the underlying control module, may lead to jerky vehicle operation.
2) The actuators (e.g., steering and braking) of AMTs are usually controlled by hydraulics, which have a large response delay (about 1.5 s) and require smoother trajectories to improve tracking performance.

Therefore, the second module refines the initial paths, ensuring curvature continuity and making local adjustments in high-risk segments to minimize collision risk and improve safety and feasibility.

IV. Methodology

A. Improved Hybrid A* Multivehicle Path Plan Method Based on Dynamic Routing Potential Fields

To preserve the efficiency of the Hybrid A* search while enabling decoupling and collision avoidance among multivehicle paths, we introduce a grid-based path potential field based on the path distributions of other vehicles into the grid search. The method for calculating the path potential field is described in Algorithm 1.

1) Path Potential Fields: To characterize the spatial distribution of vehicles and paths within the unloading area, a set of 2-D grid map points in the unloading area $\mathcal{R} \subset \mathbb{R}^2$ is defined for this area

$$\mathcal{G} = \left\{ (x_{ij}, y_{ij}) \,\middle|\, x_{ij} = x_{\min} + i \cdot \Delta x, y_{ij} = y_{\min} + j \cdot \Delta y, i, j \in \mathbb{Z} \right\} \tag{3}$$

where Δx, Δy is the resolution of the grid, $x_{\min}$, $y_{\min}$ is the minimum boundary coordinate of the grid map in the x- and y- directions, respectively, and $\mathbb{Z}$ denotes the set of integers.

In the path planning process, the dump truck travels within the unloading area, and its path P is represented as a discrete approximation of a point set on the 2-D grid map $\mathcal{G}$ as $\tilde{P} = \{(\tilde{x}_k, \tilde{y}_k)\}_{k=1}^{Q}$, where Q is the number of discrete points on the path. Each path point $(\tilde{x}_k, \tilde{y}_k)$ corresponds to the actual position of the vehicle, and considering the size and shape of the vehicle's body, a body frame polygon $\mathcal{R}_{\text{veh}}$ is formed from the path points $\tilde{P}$ to characterize the vehicle's occupied area.

For any grid point $(x_{ij}, y_{ij}) \notin \mathcal{R}_{\text{veh}}$, compute its minimum Euclidean distance to the set of interpolated path points $\tilde{P}$

$$d_{\min}(x_{ij}, y_{ij}) = \min_{(\tilde{x}_k, \tilde{y}_k) \in \tilde{P}} \sqrt{(x_{ij} - \tilde{x}_k)^2 + (y_{ij} - \tilde{y}_k)^2}. \tag{4}$$

Subsequently, the potential field values are defined using a Gaussian function. The potential field value at grid point (x_i, y_i) is given by the following equation:

$$\phi(x_i, y_i) = \phi_{\max} \exp\left(-\frac{d_{\min}(x_i, y_i)^2}{2\sigma^2} \right) \tag{5}$$

where σ is the parameter controlling the decay rate of the potential field and $\phi_{\max}$ is the maximum value of the potential field.

The following represents the overall representation of the potential field:

$$\Phi = \left\{ \phi(x_{ij}, y_{ij}) \,\middle|\, (x_{ij}, y_{ij}) \in \mathcal{G} \right\}. \tag{6}$$

The function ϕ is defined as follows:

$$\phi(x_{ij}, y_{ij}) = \begin{cases} \phi_{\max}, & \text{if } (x_{ij}, y_{ij}) \in \mathcal{R}_{\text{veh}} \\ \phi_{\max} \cdot \exp\left(-\dfrac{d_{\min}^2(x_{ij}, y_{ij})}{2\sigma^2} \right), & \text{if } (x_{ij}, y_{ij}) \notin \mathcal{R}_{\text{veh}}. \end{cases} \tag{7}$$

Moreover, to reflect the potential field values, the gradient of the potential field, defined as the gradient of the computed potential field $\nabla\phi(x_{ij}, y_{ij})$, is introduced as a corrective cost for the path direction, where

$$\nabla\phi(x_{ij}, y_{ij}) = \left(\frac{\partial\phi}{\partial x}, \frac{\partial\phi}{\partial y} \right). \tag{8}$$

In discrete grids, the gradient can be approximated using numerical methods. In this study, it is computed using the central difference method

$$\frac{\partial\phi}{\partial x} \approx \frac{\phi(x + \Delta x, y) - \phi(x - \Delta x, y)}{2\Delta x} \tag{9}$$

$$\frac{\partial\phi}{\partial y} \approx \frac{\phi(x, y + \Delta y) - \phi(x, y - \Delta y)}{2\Delta y}. \tag{10}$$

2) Improved Hybrid A Algorithm:* To enhance the Hybrid A* algorithm's ability to avoid conflicts and adaptively respond to multivehicle interactions, we integrate the previously defined grid-based path potential field into its cost function, forming an improved planner termed potential-guided Hybrid A* (PHA*).

Specifically, the potential field value $\phi(x, y)$ and its gradient $\nabla\phi(x, y)$ are incorporated into the node evaluation process. The total path cost consists of three components: the traditional path cost, $g(n)$, based on path length and turning effort; the influence of the potential field value, $\phi(x, y)$, on node attraction or repulsion; and the magnitude and directional correction of the potential field gradient, characterized by

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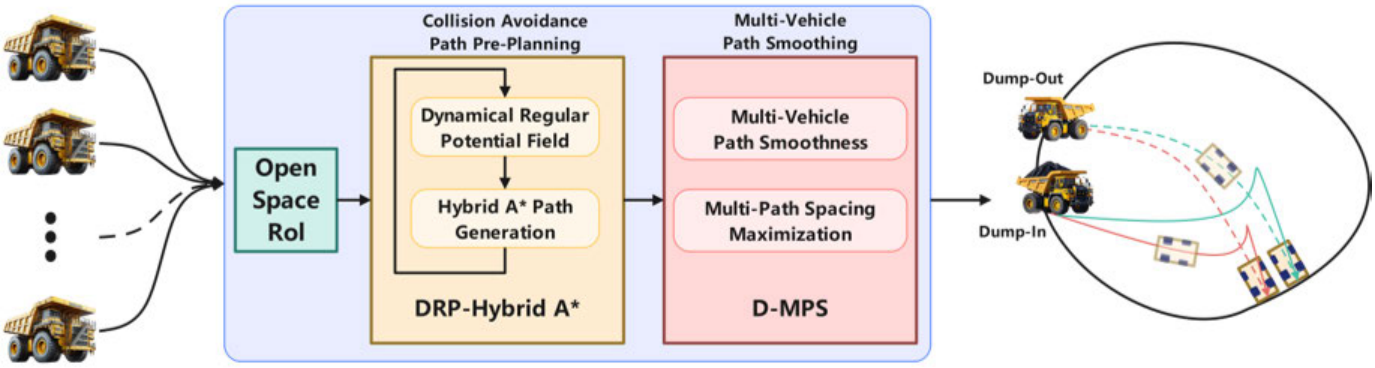


Fig. 3. Framework of proposed methodology. The first module generates decentralized, nearly collision-free truck paths. The second module smooths and refines these paths, further optimizing them to produce feasible paths that minimize collision risk.

Algorithm 1 Compute Path Potential Field

Require: G: 2D gridmap, $G = \{(x_{ij}, y_{ij})\}$;
P: Path points, $P = \{(\tilde{x_k}, \tilde{y_k})\}_{k=1}^{M}$;
$\mathcal{R}_k$: Vehicle shape polygon for path point k;
σ: Gaussian decay parameter;
$\phi_{\max}$: Maximum potential field value.
Ensure: Φ: Potential field matrix of size G.

```
Initialize Φ(x_ij, y_ij) ← 0, ∀(x_ij, y_ij) ∈ G
for all (x_ij, y_ij) ∈ G do
    if (x_ij, y_ij) ∈ ⋃_{k=1}^{M} R_k then
        φ(x_ij, y_ij) ← φ_max
    else
        d_min(x_ij, y_ij) ← min_k sqrt((x_ij − x̃_k)^2 + (y_ij − ỹ_k)^2)
        φ(x_ij, y_ij) ← φ_max · exp(−d²_min(x_ij, y_ij) / (2σ²))
    end if
    Update Φ(x_ij, y_ij) ← φ(x_ij, y_ij)
end for
return Φ
```

$|\nabla\phi(x, y)|$. Weighting parameters ω_p and ω_g are introduced to balance the contributions of the potential field values and gradients to the path planning process

$$g'(n+1) = g'(n) + \Delta d + \omega_p \cdot \phi\left(x_{ij}, y_{ij}\right) + \omega_g \cdot \left\|\nabla\phi\left(x_{ij}, y_{ij}\right)\right\|_2. \tag{11}$$

To determine suitable weight settings, we adopt a task-driven and sensitivity-based approach. In dense multivehicle scenarios emphasizing safety, higher weights (ω_p, $\omega_g = 1.5$–2.0) effectively reduce path conflicts. For efficiency-focused cases, lower values (0.5–1.0) help minimize unnecessary detours. In general scenarios, we use $\omega_p = 1.0$ and $\omega_g = 1.2$, achieving a good tradeoff between safety and efficiency.

In Hybrid A* algorithms, the heuristic value is typically calculated by combining the heuristic value from the classical A* algorithm (e.g., Euclidean or Manhattan distance) with the cost associated with the path curve (e.g., the Reed–Shepp curve). Taking the maximum of these two components ensures the consistency and admissibility of the heuristic function, thereby avoiding the underestimation of the path cost, as expressed in the following equation:

$$h(n) = \max\left\{D_{A*}(n, \text{goal}), D_{\text{RS}}(n, \text{goal})\right\}. \tag{12}$$

Since the potential value function is considered, the effect brought by the potential value function is also taken into account in the calculation of $D_{A*}(n, \text{goal})$, $D_{\text{RS}}(n, \text{goal})$, and the new formula is given as follows:

$$D'_{A*}(n, \text{goal}) \approx \sum_{i=1}^{k-1} \sqrt{(x_i - x_{i+1})^2 + (y_i - y_{i+1})^2 + \left(\omega_p \nabla\phi(x, y)\right)^2} \tag{13}$$

$$D'_{\text{RS}}(n, \text{goal}) \approx \sum_{i=1}^{k-1} \sqrt{(x_i - x_{i+1})^2 + (y_i - y_{i+1})^2 + \left(\omega_p \nabla\phi(x, y)\right)^2} \tag{14}$$

$$h'(n) = \max\left\{D'_{A*}(n, \text{goal}), D'_{\text{RS}}(n, \text{goal})\right\} \tag{15}$$

where $g(n)$ and $h(n)$ represent the cumulative cost from the past and the expected cost to the goal, respectively, and $g(n)$ consists of three components: the cost inherited from the previous node, the cost associated with the increased distance Δd, the cost due to changes in the potential field, and the potential field value at the current node. This ensures that the search is guided by both dynamic and static high-potential values while also avoiding nodes with large abrupt changes in the potential field, thereby minimizing the impact of dynamic obstacles

$$f(n) = g'(n) + h'(n). \tag{16}$$

3) Methods for the Construction of Dynamic Routing Potential Fields: To achieve efficient and decoupled path planning for AMTs in the unloading area, we propose a dynamic routing potential field (DRP) framework. The DRP is designed to adaptively guide the next vehicle's path generation by incorporating the influence of previously planned paths and unloading regions into a dynamically updated potential field. The complete planning process is described in Algorithm 2 and consists of the following steps.

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Algorithm 2 Multivehicle Path Planning With DRP-Hybrid A*

Require: Initial state D_{in}, D_{out}, dump points set $\{D_i\}$, gridmap G
Ensure: path $\mathcal{P}$

```
Initialize total path P ← ∅
Initialize total potential field Φ_total ← zeros(size(G))
for phase in [entry, exit] do
    P ← P ∪ DRP_HA*(D_in, D_out, D, Φ_total, phase, G)
end for
Function DRP_HA*
Initialize total path P ← ∅
orders ← getPlanOrders(D)
for each i in orders do
    Φ_0 ← getAdjDumpPotential(D_i)
    Φ_temp ← max(Φ_total, Φ_0)
    if phase == entry then
        P ← PHA*(D_in, D_i, Φ_temp, G)
    else
        P ← PHA*(D_i, D_out, Φ_temp, G)
    end if
    Φ_1 ← ComputePathPotentialField(P, σ, Φ_max)
    Φ_total ← max(Φ_total, Φ_1)
    P ← P ∪ P
end for
return total path P
```

1) *Unloading Point Selection:* Select the unloading point to be planned based on the principle of the unloading point being far from the entry and exit points.
2) *Initialization of the Dynamic Potential Field:* Initialize the dynamic potential field Φ_{total}, and consider the potential field Φ_0 of the unloading points adjacent to the dump point D_i.
3) *Path Planning via PHA*:* Based on the aforementioned potential field, plan the path from the first unloading point to the exit/entry point through the PHA* algorithm to generate the path potential field Φ_1.
4) *Dynamic Field Update:* After each vehicle's path is generated, update the routing potential field as

$$\Phi_{\text{total}} = \max\{\Phi_0, \Phi_1\}$$

so that subsequent vehicles are guided away from overlapping or congested regions.
5) *Repeat Steps 1–4 for All Remaining Vehicles:* The DRP evolves with each vehicle's path, dynamically guiding the planning of subsequent vehicles in a decoupled and adaptive manner.

B. Collision Risk Minimization Multivehicle Path Smoothing Optimization Algorithm

The DRP-Hybrid A* planning result is M paths containing the entry/exit D_{in}–D_i–D_{out} paths, and to further ensure the smoothness of the paths with the improvement of the collision-free probability between the paths, it is necessary to optimize the DRP-Hybrid A* planning results.

1) Initial Path Preprocessing: The initial paths generated by DRP-Hybrid A*, particularly those involving reverse parking maneuvers, require endpoint correction to ensure heading consistency. This preprocessing is carried out using the following formulations:

$$\mathcal{T}_n = \{k \,||\theta_{i+1} - \theta_i| \approx \pi, k = 0, \ldots, n_e - 1\} \tag{17}$$

$$\varphi_n(i) = \begin{cases} \theta_n(i), & \text{if } i \in \mathcal{T}_n \\ (\theta_n(i) + \pi) \bmod 2\pi, & \text{if } i \notin \mathcal{T}_n \end{cases} \tag{18}$$

where θ_i is the heading angle of the path P_n and $\mathcal{T}_n$ denotes the set of heading reversal points. The corrected start and end headings of the path are denoted by $\varphi_n(0)$ and $\varphi_n(n_e)$, respectively. The unit direction vector corresponding to $\varphi_n(i)$ is given by $\hat{\mu}_{\varphi_n(i)}$. Let $p_n(i) = (x_n(i), y_n(i))$ denote the 2-D coordinates of the ith point along P_n. To implicitly encode the desired terminal heading during smoothing, the second-to-last point of the path is adjusted in the direction of $\varphi_n(n_e)$, ensuring that the smoothed trajectory is aligned with the desired final heading

$$p_n(1) = p_n(0) + \|p_n(1) - p_n(0)\| \cdot \hat{\mu}_{\varphi_n(0)} \tag{19}$$

$$p_n(n_e - 1) = p_n(n_e) + \|p_n(n_e - 1) - p_n(n_e)\| \cdot \hat{\mu}_{\varphi_n(n_e)}. \tag{20}$$

2) Objective Function: The total objective function consists of two parts: minimizing path roughness and optimizing spatial separation, which are mathematically formulated as follows:

$$\min \mathcal{J} = \omega_{\text{rough}} \cdot \sum_{n=1}^{N} J_{\text{rough}}(P_n) + \omega_{\text{spacing}} \cdot \sum_{i=1}^{N} \sum_{j=1, j\neq i}^{N} J_{\text{spacing}}(P_i, P_j) \tag{21}$$

where ω_{rough} and ω_{spacing} are the path roughness weight and the path spacing weight, respectively. $J_{\text{rough}}(P_n)$ represents the roughness function of the nth path. For the path p_n, its roughness can be measured by the second-order difference (curvature approximation) of each point

$$J_{\text{rough}}(P_n) = \sum_{i=1, i\notin\mathcal{T}_n}^{n_e-1} \|2p_n(i) - p_n(i-1) - p_n(i+1)\|^2 \tag{22}$$

where $\|\cdot\|$ denotes the Euclidean norm.

To penalize overcrowded paths, a path spacing cost function J_{spacing} is designed to measure the rationality of the spatial distribution between paths. This function uses an exponential decay model to characterize the impact of the distance between path points on the cost

$$J_{\text{spacing}}(P_i, P_j) = \sum_{k=0}^{i_e} \sum_{l=0}^{j_e} \exp\left(-\frac{\|p_i(k) - p_j(l)\|}{s_{\min}}\right) \tag{23}$$

where $J_{\text{spacing}}(P_i, P_j)$ represents the spacing function between paths i and j. $p_i(k)$ represents the horizontal and vertical coordinates of the kth point on the ith route. $p_j(l)$represents the horizontal and vertical coordinates of the lth point on the jth route. $s_{\min}$ denotes the minimum safe spacing threshold

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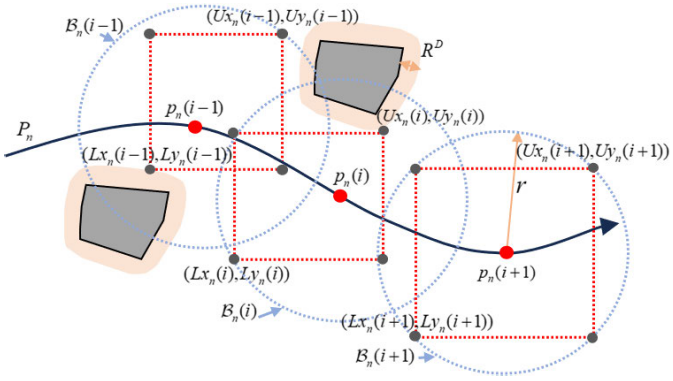


Fig. 4. Generation of box constraints: obstacles are inflated by R_D, and the vehicle is treated as a point. A feasible region with radius r is defined around each path point, and the largest rectangle tangent to the inflated obstacles is extracted to determine the bounds $[(Lx_n(i), Ly_n(i)), (Ux_n(i), Uy_n(i))]$.

between paths P_i and P_j. $\exp(\cdot)$ represents the natural exponential function.

3) Adjustable Box Constraints: To ensure obstacle avoidance during path optimization, a spatial corridor is constructed along the reference path. For each path point i, a circular safety region $\mathcal{B}_i$ of radius r is defined.

To simplify collision checking, the vehicle is modeled as a point by inflating obstacles with a radius

$$R_D = \frac{1}{2}\sqrt{\left(\frac{l}{2}\right)^2 + w^2} \tag{24}$$

where l and w are the vehicle's length and width.

Within each $\mathcal{B}_i$, a maximum axis-aligned rectangle $\mathcal{U}_i \subseteq \mathcal{B}_i \setminus \mathcal{O}_D$ is extracted by solving

$$\max_{x_1,y_1,x_2,y_2} \quad (x_2 - x_1)(y_2 - y_1) \tag{25}$$

$$\text{s.t.} \quad (x_1, y_1), (x_2, y_2) \in \mathcal{B}_i \setminus \mathcal{O}_D. \tag{26}$$

The set of rectangles $\{\mathcal{U}_i\}$ defines the spatial corridor, where each $\mathcal{U}_i$ represents the local feasible optimization region for the ith path point. Here, $\mathcal{O}_D$ denotes the inflated obstacle region obtained by expanding each obstacle by radius R_D to account for the vehicle's dimensions. These regions are bounded by upper limits U_{x_k}, U_{y_k} and lower limits L_{x_k}, L_{y_k}, as illustrated in Fig. 4. The corresponding box constraints are formulated as

$$Lx_n(k) \le x_n(k) \le Ux_n(k), \quad k = 1, \ldots, n_e - 2 \tag{27}$$

$$Ly_n(k) \le y_n(k) \le Uy_n(k), \quad k = 1, \ldots, n_e - 2. \tag{28}$$

The index range $k = 1, \ldots, n_e - 2$ excludes the start and end control points of the trajectory, applying constraints only to intermediate points subject to optimization.

4) Curvature Constraint: For the kth point on the nth path, the curvature constraint is expressed as

$$\|2p_n(k) - p_n(k-1) - p_n(k+1)\|^2 \le \left(\Delta s^2 \cdot \mathcal{K}_{\max}\right)^2 + \varepsilon_n^k, \quad k = 1, \ldots, n_e - 2 \tag{29}$$

where $\mathcal{K}_{\max}$ denotes the maximum allowable path curvature, determined by the vehicle's minimum turning radius $r_{\min}$. To improve feasibility and convergence, a slack variable ε_n^k is introduced, allowing limited deviations from the strict curvature bound when necessary. This ensures that trajectories remain within the vehicle's kinematic limits while maintaining flexibility for practical optimization.

5) Optimization Problem: This optimization problem comprehensively considers path smoothness, spatial separation between paths, and physical constraints. It ensures trajectory feasibility through spatial corridor constraints and curvature constraints while minimizing path roughness and maximizing spatial separation between paths

$$\min \mathcal{J} = \omega_{\text{rough}} \cdot \sum_{n=1}^{M} (22) + \omega_{\text{spacing}} \cdot \sum_{n1=1}^{M} \sum_{n2 \ne n1}^{M} (23) \tag{30}$$

$$\text{s.t. } (27), (28), (29)$$

$$p_n(0) = p_n^{\text{ref}}(0), \quad p_n(n_e) = p_n^{\text{ref}}(n_e), \quad n = 1, \ldots, N. \tag{31}$$

The optimization problem is modeled using MATLAB's YALMIP framework, where path variables are uniformly encoded to construct a nonlinear optimization model, which is efficiently solved using the fmincon solver to obtain the optimal paths.

V. Simulation and Experiment

To comprehensively evaluate the performance of DRP-Hybrid A*, CRM-MVPO, and their combined algorithm, this section designs three sets of comparative experiments to assess path dispersion, trajectory safety, and replanning capabilities. All experiments were conducted on an AMD Ryzen 7 4800U platform using the MATLAB framework. First, in a multivehicle setting, we compare DRP-Hybrid A* against the Hybrid A* from the MATLAB Navigation Toolbox to observe differences in path dispersion and collision risk. Next, on identical planned routes, we benchmark CRM-MVPO alongside Baidu Apollo's FEM POS Deviation Smoother and a CBS-based planner (which only handles planning for different start-goal pairs) to assess the collision-free distance proportion and path smoothness. Finally, in a complex unloading environment, we test the full DRP-Hybrid A* + CRM-MVPO system's replanning and deadlock-avoidance capabilities against the CBS planner.

The experiments adopt the spatial collision-free distance proportion, η, as the primary evaluation metric, defined as the proportion of a vehicle's planned path that adheres to the minimum safety distance constraint during multivehicle collaborative operations. Supplementary metrics, including path length L and computation time T, are assessed.

The collision-free distance proportion of a single path is calculated as

$$\eta_n = \frac{s_{n,\text{collision}_{\text{free}}}}{s_n} \tag{32}$$

where s_n is the length of path n, computed from the sum of Euclidean distances between waypoints, and $s_{n,\text{collision}_{\text{free}}}$ is the length of collision-free segments considering the truck's body footprint.

We introduce the following key assumptions at the start of our experiments.

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TABLE I
LIST OF SIMULATED PARAMETER SETTINGS

Parameter	Description	Value
N	Number of AMTs	2-5
l	The length of the truck	15.6 m
w	The width of the truck	8.69 m
l_{wb}	Rear axle to front distance of the truck	10.82 m
$r_{\min}$	Minimum turning radius	14.85 m
l_m	Motion primitive length	6.0 m
n_m	Number of motion primitives	5
$RS_{interval}$	RS planning interval	50
ϕ_{max}	Max potential field value	2
σ	Potential field decay factor	2
ω_p	Weight of potential field	1.0
ω_g	Weight of potential field gradient	1.2
ω_{smooth}	Path smoothness weight	1.0
$\omega_{spacing}$	Path spacing weight	1000.0
δ_s	Path point spacing	3.0 m
r	Safety region radius for each path point	12.0 m
κ_{max}	Max curvature of path	0.067
s_{min}	Min spacing between paths	1.0
ε_k	Curvature relaxation term	0.02

1) The unloading-area map (static obstacles, road boundaries, and dump points) is known a priori and remains fixed.
2) All AMTs are homogeneous with identical kinematic/dynamic limits ($v_{\max}$, $a_{\max}$, and $r_{\min}$) and perfect trajectory tracking.
3) *Localization Is Exact:* No sensor or perception models are required.
4) Unloading-point service orders and intervehicle time synchronization are provided by an upstream scheduler, with no time drift or communication delays considered in this study.

The experiments are based on the unloading area of the Haluusu open-pit mine in Inner Mongolia Autonomous Region of China, reflecting realistic operating conditions and vehicle characteristics. Key planning parameters, such as motion primitives and RS curve iterations in the Hybrid A* algorithm, are kept fixed. The parameters for the simulation experiments are listed in Table I.

A. Case 1

In this experiment, the AMTs will enter from the same entry point and exit from the same exit point. We have selected dump points 8 and 10 in the unloading area, and the AMTs will travel to these points to perform the unloading task. The Hybrid A* and DRP-Hybrid A* planners generate the paths based on the selected dump points.

The planning results are shown in Fig. 5. The left display shows the planning result of the proposed method, while the right displays the result of the Hybrid A* method. Compared to Hybrid A*, the proposed method produces more spatially distributed paths.

The dynamic potential field after the planning of the first entry path is shown in Fig. 6. The red entry path is influenced by this potential field, searching in the direction of lower values of the potential field, resulting in a more decentralized path. Table II shows the percentage of the distance traveled without a potential collision point for each path. Compared to Hybrid A*, our method shows a significant advantage in

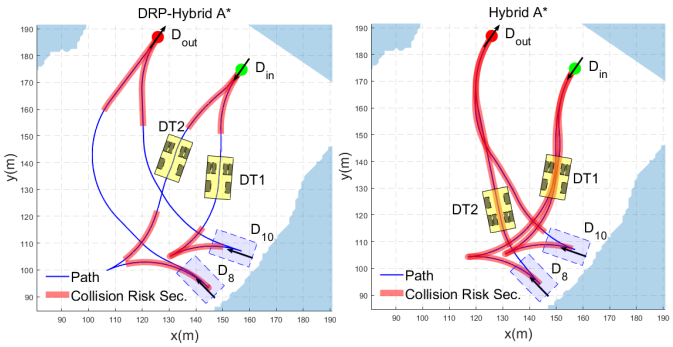


Fig. 5. Comparison of planning results between Hybrid A* and DRP-Hybrid A* methods: green and red points represent the entry and exit points, respectively. DT1 and DT2 denote the operating dump trucks, which travel from the entry point D_{in} to the unloading points D_{10} and D_8 and then exit from the exit point D_{out}.

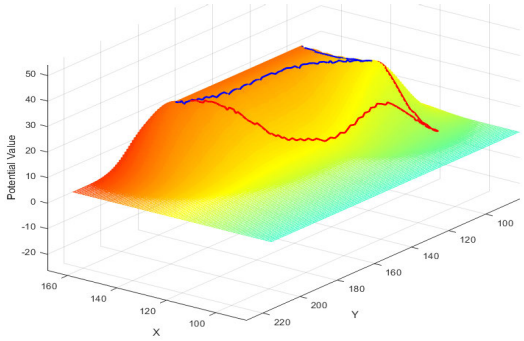


Fig. 6. Path planning results under the influence of the path potential field. The red path deviates due to the potential field generated by the blue path.

collision-free distance proportion, with only a slight increase in path length and computation time.

To evaluate our algorithm's performance under varying vehicle counts and unloading point combinations, we assess three metrics: average collision-free distance proportion ($\bar{\eta}$), average path length ($\bar{L}$), and average computation time per path ($\bar{T}$).

We test combinations of two to five dump points selected from 10, excluding adjacent ones per operational rules, with each truck selecting one dump point. For each combination, both Hybrid A* and DRP-Hybrid A* generate entry and exit paths, which are then evaluated using the above metrics.

The average collision-free distance proportion is computed as

$$\bar{\eta} = \frac{\sum_{i=1}^{N} \sum_{j=1}^{C} \left(\eta_{\text{entry},ij} + \eta_{\text{exit},ij}\right)}{2C \cdot N} \tag{33}$$

where C is the number of valid unloading combinations (36, 56, 35, and six for two, three, four, and five AMTs, respectively), N is the number of AMTs, and $\eta_{\text{entry},ij}$ and $\eta_{\text{exit},ij}$ denote the proportion of collision-free distances along the entry and exit paths of truck j under combination i. $\bar{L}$ and $\bar{T}$ are computed in the same manner by replacing η with the corresponding path length L and computation time T terms.

The experimental results demonstrate that the proposed DRP-Hybrid A* method consistently outperforms the standard Hybrid A* in terms of collision-free distance across

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TABLE II
COMPARISON OF COLLISION-FREE DISTANCE PROPORTION AND PATH LENGTH OF TWO METHODS

Method	Collision-Free Dist. Prop. ($\bar{\eta}$, %)				Path Length(L, m)		Computation Time(T, s)	
	$P_{entry,1}$	$P_{entry,2}$	$P_{exit,1}$	$P_{exit,2}$	P_{entry}	P_{exit}	P_{entry}	P_{exit}
Hybrid A*	0.95	0.00	12.55	17.08	224.58	206.32	0.3243	0.1879
DRP-Hybrid A*	39.62	34.23	63.98	70.49	251.65	210.51	0.9687	0.5179

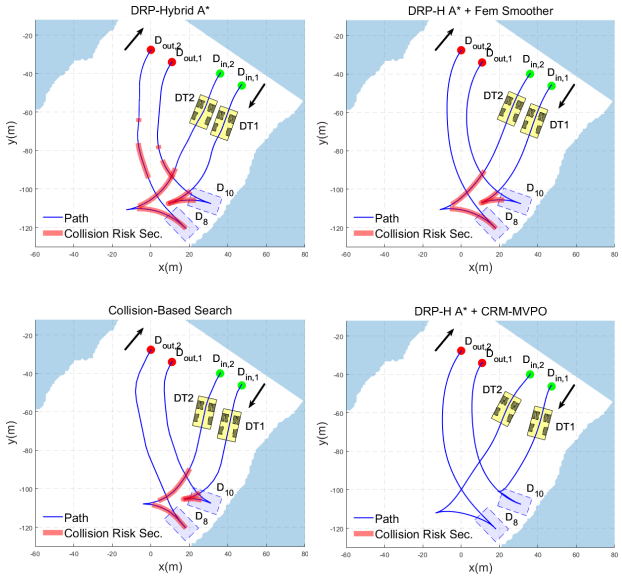


Fig. 7. Comparison of different methods for generating the entry/exit paths of two AMTs: green and red points indicate entry and exit locations. DT1 and DT2 travel from their respective entry points $D_{\text{in},1}$ and $D_{\text{in},2}$ to D_{10} and D_8 and then exit from $D_{\text{out},1}$ and $D_{\text{out},2}$.

different numbers of trucks. As shown in Table III, the average collision-free distance proportion on full paths increases from 42.23% to 64.17% in the two-truck case, marking a 52% relative improvement. For three to five trucks, the improvement rates remain high at 94.7%, 89.7%, and 84.5%, respectively.

Although DRP-Hybrid A* introduces moderate increases in path length (e.g., from 150.88 to 159.95 m in the two-truck case) and computation time (e.g., from 0.324 to 0.969 s), the safety and coordination benefits clearly outweigh these costs.

In summary, DRP-Hybrid A* enhances collision avoidance and path flexibility, with improvements becoming more significant as the traffic density increases.

B. Case 2

In this experiment, the unloading area has multiple entry and exit points. The AMTs follow paths planned according to different entry and exit points and dump points, selecting the unloading routes $D\text{in}, 1 - D_{10} - D\text{out}, 1$ and $D\text{in}, 2 - D_8 - D\text{out}, 2$.

The results of different methods are shown in Fig. 7. DRP-Hybrid A* can quickly plan feasible paths, but collision risk still exists near the entry changeover point and in the mid-section of the exit path. Baidu Apollo Fem Smoother improves curvature and smoothness but does not consider interactions between vehicles, so collision issues remain. CBS eliminates spatiotemporal conflicts through conflict detection and priority assignment, yet spatial collision risks still persist.

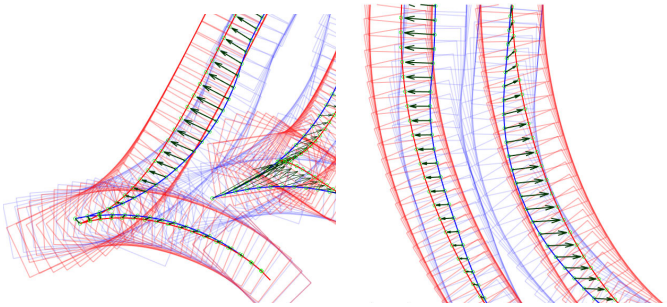


Fig. 8. Results after applying CRM-MVPO optimization to DRP-Hybrid A*: blue and red rectangles represent the vehicle path projections before and after optimization, respectively. Arrows indicate the displacement of corresponding path points.

The proposed method first uses DRP-Hybrid A* to generate dispersed paths and then optimizes trajectories with CRM-MVPO, successfully avoiding collision risks on both entry and exit paths.

As shown in Fig. 8, the truck spacing diagrams for the entry and exit paths before and after CRM-MVPO, considering the vehicle's body frame, are presented. Before optimization, both paths have sections where the collision lines intersect. After optimization, the paths remain within the collision safety zone, and the proportion of collision-free distance for the entry and exit paths improves from 82.96% and 77.43%, respectively, to 100%. Similarly, the collision-free distance proportion for exit paths increases from 70.40% and 52.66% to 100%.

Table IV compares methods on collision avoidance, path quality, and computation time. Hybrid A* is fastest (1.269 s) but least safe (28.35%, 28.89%). CBS improves safety to (66.04%, 55.72%) but with a higher cost (12.503 s). DRP-Hybrid A* balances safety (39.57%, 37.41%) and runtime (2.160 s), with over 70% collision-free entry paths. Combining it with Fem Smoother ensures 100% safe exits though overall safety is uneven. Hybrid A* + CRM-MVPO greatly enhances safety (66.72%, 54.35%) but requires a much longer time (32.030 s). The best results come from DRP-Hybrid A* + CRM-MVPO, achieving 100% collision-free paths and the highest overall safety (68.13%, 55.91%).

Fig. 9 shows the entry and exit path curvatures of both trucks before and after CRM-MVPO optimization. In the exit path, minor collision risks are mitigated by smoothing the curvature. In the entry path, path separation is adjusted within the curvature limit to avoid collisions.

C. Case 3

In order to test the ability of the proposed method to handle complex working conditions, this experiment involves five AMTs in the unloading area. One truck is faulty and cannot move, two fully loaded AMTs (DT1 and DT2) will

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TABLE III
PERFORMANCE COMPARISON OF HYBRID A* AND DRP-HYBRID A* UNDER DIFFERENT NUMBERS OF TRUCKS

Method	Number of Trucks	Avg. Collision-Free Dist. Prop. ($\bar{\eta}$, %)		Avg. Path Length ($\bar{L}$, s)		Average Computation Time ($\bar{T}$, s)	
		$\bar{P}_{entry}$	$\bar{P}_{exit}$	$\bar{P}_{entry}$	$\bar{P}_{exit}$	$\bar{P}_{entry}$	$\bar{P}_{exit}$
Hybrid A*	2	32.5	51.95	150.883	128.261	0.324	0.188
	3	16.60	39.29	150.921	128.443	0.314	0.187
	4	11.12	31.33	150.986	128.561	0.324	0.189
	5	7.65	25.01	151.075	128.479	0.363	0.197
DRP-Hybrid A*	2	70.01	58.33	159.946	130.663	0.969	0.518
	3	57.39	51.39	161.673	132.969	1.042	0.495
	4	39.02	41.54	169.683	134.924	1.370	0.494
	5	23.61	36.66	175.379	135.635	1.733	0.528

TABLE IV
COMPARISON OF COLLISION-FREE DISTANCE PROPORTION AND PATH LENGTH OF EACH METHOD

Method	Entry Path(η, %)		Exit Path(η, %)		Overall Path(η, %)		Total Computation Time(T, s)
	$P_{entry,1}$	$P_{entry,2}$	$P_{exit,1}$	$P_{exit,2}$	P_1	P_2	
Hybrid A*	27.99	25.99	45.52	49.44	28.35	28.89	1.269
DRP-Hybrid A*	70.40	52.66	82.96	77.43	39.57	37.41	2.160
Conflict-Based Search	72.60	50.29	100	100	66.04	55.72	12.503
DRP-Hybrid A* + Fem Smoother	74.35	53.74	100	100	48.59	45.65	13.213
Hybrid A* + **CRM-MVPO**	87.99	82.91	100	100	66.72	54.35	32.030
DRP-Hybrid A* + CRM-MVPO	100	100	100	100	68.13	55.91	32.305

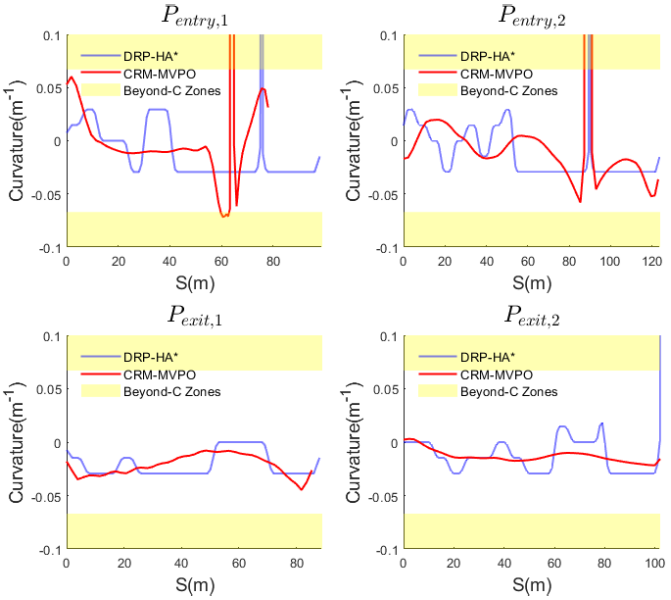


Fig. 9. Curvature of entry and exit paths: the yellow area indicates regions exceeding the maximum curvature limit, with curvature values approaching infinity at direction-switching points due to sudden changes.

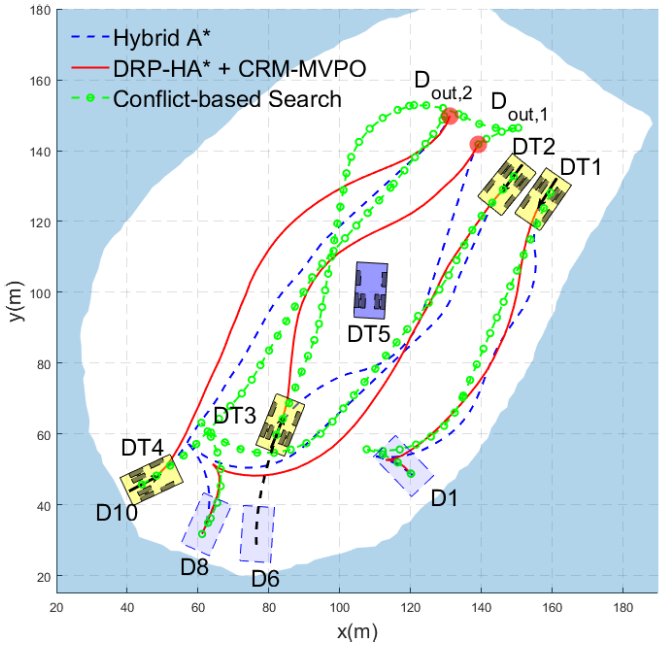


Fig. 10. Results of multivehicle planning: DT1 is directed to D_{10}, DT2 to D_8, DT3 to $D_{\text{out},1}$, DT4 to $D_{\text{out},2}$, and DT5 is disabled due to a malfunction.

go to the dump point to perform the unloading task, one dump truck (DT4) is unloading, and one unloaded dump truck (DT3) has already left the dump point and is heading to the exit. The proposed method will plan the respective paths for these vehicles to avoid obstacles, reduce the collision risk, and ensure the smoothness of the paths.

Fig. 10 shows the planning results, with each truck's path labeled by its number. The blue dashed line represents the Hybrid A* results, where paths 2 and 3 significantly overlap, and potential collisions occur at switchover points between paths 4 and 2, and paths 1 and 2. The green dashed circles (CBS) show large detours, such as truck 3's path, reflecting conservative avoidance at the cost of efficiency. The solid red line represents the results of the proposed method. Taking into account obstacles and the potential field of route 2, the method adjusts paths 3 and 4 to the left of the faulty truck and further separates path 4 from path 3. Paths 1 and 2 are also adjusted at the switchover point to avoid collisions.

As shown in Table V, DRP-Hybrid A* + CRM-MVPO improves the collision-free distance proportion to 86.06% and shortens the path length, albeit with increased computation time, indicating a favorable tradeoff between planning quality and efficiency.

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TABLE V
PERFORMANCE COMPARISON OF HYBRID A* AND DRP-HYBRID A* UNDER DIFFERENT NUMBERS OF TRUCKS

Method	$\bar{\eta}(\%)$	L(s)	T(s)
Hybrid A*	47.31	130.196	0.395
Conflict-Based Search	54.03	141.25	4.378
DRP-Hybrid A* + CRM-MVPO	86.06	124.972	7.515

VI. CONCLUSION

In this article, we propose an innovative multivehicle collaborative path planning method for minimizing collision risk. This method can plan paths with the smallest possible or even completely collision-free probability for multiple AMTs operating simultaneously in unstructured scenarios, such as unloading areas. In addition, the method further improves the paths' collision probabilities and smoothness through a multivehicle path optimizer. To evaluate the performance, we design a series of multivehicle operation planning simulations for the unloading area scenario and assess the spatial collision-free distance proportion of each path after planning and optimization. The results demonstrate that our method ensures path smoothness while significantly reducing the multivehicle spatial collision probability. Therefore, the proposed method can serve as a reference to improve the operational safety of AMTs in unloading areas and similar work scenarios, thus enhancing the overall economic benefits of AMTs.

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